

Agenda and Minutes 23-05-25

☐ Meeting Agenda – *Team 13 - 23-05-25*

Group Number: 13 **Date:** 23-05-25

Time: Start - End

Location: [Room/Platform]

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
 - Hristiyan Georgiev (241226)
 - Mohammadali Jaber (244437)
 - Aristotelis Protopapas (234062)
-

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Details of the project including data sources, feasibility, and expected inputs and outputs of the model.	Discussion, no presenter	20min	
2	Timeline of the project	Discussion, no presenter	10min	

☐ Meeting Minutes – *Team 13 - 23-05-25*

Group Number: 13 **Date:** 23-05-25

Time: 10:00 - 10:30

Location: On Campus

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) - Present
 - Hristiyan Georgiev (241226) - Present
 - Mohammadali Jaber (244437) - Absent - Was filling sick, informed the team later in the day.
 - Aristotelis Protopapas (234062) - Absent - No show, no contact with the team.
-

□ Discussion Summary

Agenda item 1: Details of the project including data sources, feasibility, and expected inputs and outputs of the model

Key Discussions - agenda item 1

- Feature Engineering Process & Resulting Dataset.
- Target Variable Definition (event column)

Decisions Made - agenda item 1

- The project will proceed using `AAPL_consolidated_features_events_final.csv` as the primary dataset for model development.
- The enhanced multi-class event detection logic and the comprehensive feature set (including initial technicals, advanced time-based, VoV, and interaction features) are finalized for this stage.
- The data engineering phase is considered complete, with the dataset ready for pre-modeling preparation (feature shifting, splitting).

Agenda item 2: Timeline of the project

Key Discussions - agenda item 2

- Hristiyan at the beginning (week 3) will be responsible for initial EDA, Data Collection, and Pre-processing, as he has been researching that topic for 3 weeks now and has some initial findings. After creating the initial dataset he will send it to Szymon and Mohammadali for modelling. In later weeks he will still work with collecting data and potentially implementing NLP, and if needed he will help the modeling team with the ML development.
- Szymon will take care of planning and distributing the team resources throughout the project timeline. He will be responsible for creating ML models, a GitHub repository, and coordinating the work if needed.
- Mohammadali will be responsible for creating the supporting ML models (e.g. when the team will be testing multiple options) and XAI part of the project. Additionally, he will help Hristiyan with Data Collection and Pre-processing, although the details will be discussed between them and added to the Trello.
- Aristotelis will be responsible for the dashboard/front-end part of the application ensuring compatibility with the client's requirements and current industry UX/UI design standards.

Decisions Made - agenda item 2

- Resolutions or Agreements: As stated above. Moreover, Szymon will prepare Task 2.2 by himself, as the main idea behind the group project remained mainly unchanged from his individual proposal, so only minor changes in the task content are needed.
 - Unresolved Issues: Half of the team was absent, thus it was impossible to create a detailed plan and task division, this topic will be discussed at Monday's team meeting.
-

☐ Action Items

Task	Responsible	Deadline
Preparing the data sources overview	Hristiyan	27.05 9 AM
Create task distribution for week 3	Szymon	26.05 9 AM

☐ Make sure all tasks and deadlines are also logged in Trello or your task manager.

☐ Next Meeting

Date: 26-05-2025

Time: 20:00

Location: Teams Call

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621)

Agenda and Minutes – 26-05-2025

☐ Meeting Agenda – Team 13 – 26-05-2025

Group Number: 13 **Date:** 26-05-2025 **Time:** 20:00 **Location:** Teams Call **Chair:** Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaberli (244437)
- Aristotelis Protopapas (234062)

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Briefing team members who missed the last retrospective on the project direction and high-level task allocation	Szymon, Hristiyan	10 min	

#	Topic	Presenter	Duration	Notes
2	Discussion on the remaining sections of the group project proposal that need to be completed	Szymon	10 min	
3	Review of repository contribution rules and the CONTRIBUTING.md file	Szymon	5 min	

☐ Meeting Minutes – *Team 13* – 26-05-2025

Group Number: 13 **Date:** 26-05-2025 **Time:** 20:00 – 20:20 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
 - Hristiyan Georgiev (241226) – Present
 - Mohammadali Jaberli (244437) – Present
 - Aristotelis Protopapas (234062) – Present
-

☐ Discussion Summary

Agenda Item 1: Retrospective Briefing

Key Points:

- Team members were updated on key decisions from the last retrospective (see Meeting Minutes – 23-05-2025). No objections were raised regarding the project's direction or task assignments.

Note: When asked about his progress on the UX/UI design research task, Aristotelis did not respond. This could have been due to a technical issue. He will be asked again during Tuesday's DataLab session.

Decisions:

- No formal decisions were made.
-

Agenda Item 2: Project Proposal Review

Key Points:

- Szymon shared the parts of the group project proposal he has completed.
- Hristiyan was assigned to complete the sections related to data, as he is responsible for that component of the project.
- The "Project Timeline" section still needs to be finalized.

Decisions:

- Hristiyan will complete his assigned sections by Wednesday morning.
- The entire team will work on the “Project Timeline” during Tuesday’s session.

Agenda Item 3: Repository Guidelines

Key Points:

- Szymon explained the proper use of the repository and introduced the `CONTRIBUTING.md` file as a reference.
- The team reviewed and accepted the guidelines.

Decisions:

- If any member is unsure about a task or encounters an issue, they are to consult Szymon for clarification.

☐ Action Items

Task Responsible Deadline

N/A N/A N/A

- ☐ *Ensure all tasks and deadlines are recorded in Trello or your task management tool.*

☐ Next Meeting

Date: 27-05-2025 **Time:** 10:00 – 10:30 **Location:** On Campus **Chair:** Szymon Chirowski (242621),
Hristiyan Georgiev (241226) **Minutes Taker:** Hristiyan Georgiev (241226)

Agenda and Minutes 28-05-2025

☐ Meeting Agenda – *Team 13 - 28-05-2025*

Group Number: 13 **Date:** 28-05-2025 **Time:** 17:00 – 17:30 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)

- Hristiyan Georgiev (241226)
- Mohammadali Jaberī (244437)
- Aristotelis Protopapas (234062)

□ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Data exploration and processing conclusions: what we have, what we don't	Hristiyan	10 min	Overview of current state of data
2	Discussion on initiating the first model builds	Szymon	10 min	Assess readiness to start modeling
3	Task division for the upcoming week	Team	5 min	Clarify who does what next
4	Progress update on Aristotelis' task	Aristotelis	5 min	UX/UI research task follow-up

□ Meeting Minutes – *Team 13 - 28-05-2025*

Group Number: 13 **Date:** 28-05-2025 **Time:** 17:00 - 17:15 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
 - Hristiyan Georgiev (241226) – Present
 - Mohammadali Jaberī (244437) – Present
 - Aristotelis Protopapas (234062) – Present
-

□ Discussion Summary

Agenda item 1: Data exploration and processing conclusions

Key Discussions

- Hristiyan described the current dataset combining minute-by-minute and daily Apple stock data.
- Excel format is used due to CSV formatting issues.
- Grouping events by 30-minute intervals is planned, but shorter intervals (15, 10, 5 mins) will also be explored.
- Awaiting modeling feedback before integrating external datasets.

Decisions Made

- Continue working with Excel for now.
- Delay merging external datasets until current dataset is validated.
- Test multiple time window granularities for event prediction.

Unresolved Issues

- Conversion of dataset from Excel to properly formatted CSV without corrupting number formats.
 - Final decision on optimal time window for event grouping.
-

Agenda item 2: Discussion on initiating the first model builds

Key Discussions

- Mohammadali began working on a random classifier and is reviewing dataset structure.
- Planning to experiment with short-term memory (LSTM) models.
- Szymon will try basic classifiers for comparison.
- Discussion about limiting the scope to Apple stock for initial model validation.

Decisions Made

- Continue with both LSTM and traditional classifiers.
- Share performance metrics and models for peer review.
- Focus solely on Apple stock initially for proof-of-concept modeling.

Unresolved Issues

- Final model approach still undecided; depends on performance results.
-

Agenda item 3: Task division for the upcoming week

Key Discussions

- Mohammadali will document planned modeling tasks in Trello.
- Hristiyan will delay further data engineering until feedback is received but is available to help with model execution.
- Szymon will test simpler models and provide feedback.

Decisions Made

- Update Trello with all individual responsibilities and task statuses.
- Cross-support between modeling and data processing if needed.

Unresolved Issues

- Clarity on next dataset engineering steps pending modeling results.

Agenda item 4: Progress update on Aristotelis' task

Key Discussions

- Aristotelis created a draft document detailing UI/UX best practices and technology stack.
- The draft will be shared for team review via Teams.
- Estimated final version by Monday, possibly sooner.

Decisions Made

- Upload the draft to Teams for review.
- Submit final version by 02-06-2025.

Unresolved Issues

- Pending team feedback on the document.
- Final adjustments may be required based on review.

☐ Action Items

Task	Responsible	Deadline
Upload initial modeling results and performance summary	Mohammadali	29-05-2025
Share draft UI/UX stack and design guideline doc	Aristotelis	30-05-2025
Review and provide feedback on UI/UX doc	All	30-05-2025
Experiment with classic classifiers as baseline	Szymon	03-06-2025
Log individual tasks and updates in Trello	All	Ongoing

☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

Date: 02-06-2025 **Time:** TBD **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:**
Szymon Chirowski (242621)

Agenda and Minutes

27/05/2025

□ Meeting Agenda – *Team 13* - 27/05/2025

Group Number: 13

Date: 27/05/2025

Time: 09:58 – 10:40

Location: Teams Meeting

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621)

Attendees:

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaberli (244437)
- Aristotelis Protopapas (234062)
- Elavendan Rajendran

□ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Retrospective Recording & Attendance	Elavendan	5 mins	Discussed recording requirements and importance of attendance.
2	News Event Data Review	Elavendan, Hristiyan	10 mins	Issues with data formatting, reliability (broken links), and structure.
3	Project Idea Clarification (Event Prediction vs. Outlier Removal)	All	5 mins	Confirmed primary goal is event prediction, news sentiment is optional.
4	Input/Output Granularity & Event Definition	All	5 mins	Minute-level data, predicting 5 event types in a 30-min window.
5	Project Timeline & Deliverables	Elavendan	3 mins	EDA, final table, model building, backtesting, UI, submission deadlines.
6	Backtesting Strategy	Elavendan	3 mins	Explained approach with cutoff dates.

#	Topic	Presenter	Duration	Notes
7	EDA Review (Hristiyan's work) & Data Validation	Hristiyan, Elavendan	7 mins	Reviewed features, identified potential data issues (e.g., large price jumps).
8	Meeting Minutes Template	Szymon, Hristiyan	2 mins	Agreed to use Markdown template instead of Word.
9	Task Distribution & Trello Documentation	Szymon	2 mins	Discussed roles and need to document in Trello.

☐ Meeting Minutes – *Team 13* - 27/05/2025

Group Number: 13

Date: 27/05/2025

Time: 09:58 – 10:40

Location: Teams Meeting

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621)

Attendees:

- Szymon Chirowski (242621) – on time
- Hristiyan Georgiev (241226) – on time
- Mohammadali Jaber (244437) – on time
- Aristotelis Protopapas (234062) – on time
- Elavendan Rajendran – on time

☐ Discussion Summary

Agenda item 1: Retrospective Recording & Attendance

Key Discussions - agenda item 1

- Need to record sessions for evidencing.
- Mohammadali and Aristotelis warned about missing retrospectives and impact on grades.

Decisions Made - agenda item 1

- Future retrospectives must be attended by all.
- Recordings or documentation are needed for evidencing.

Agenda item 2: News Event Data Review

Key Discussions - agenda item 2

- Data is unstructured, broken, and links may be unreliable.
- Sentiment polarity scores might not be used.
- Hugging Face model proposed for sentiment.

Decisions Made - - agenda item 2

- News event data is optional and not ready for use.
- If used, a reliable source is needed.

Unresolved Issues - agenda item 2

- Data source reliability and formatting.
-

Agenda item 3: Project Idea Clarification (Event Prediction vs. Outlier Removal)

Key Discussions - agenda item 3

- Confusion on predicting events vs. removing outliers.
- Primary goal: event prediction from last retrospective.
- News sentiment is optional.

Decisions Made - agenda item 3

- Project will focus on predicting specific market events.
 - News data is optional.
-

Agenda item 4: Input/Output Granularity & Event Definition

Key Discussions - agenda item 4

- Predicting 5 event types.
- Window: next 30 minutes using minute-level data.

Decisions Made - agenda item 4

- Predict specific event types within a 30-minute window.
 - Use defined event categories.
-

Agenda item 5: Project Timeline & Deliverables

Key Discussions - agenda item 5

- EDA by this week.
- Final table by next Tuesday.
- Model building week 5.
- Backtesting & UI week 6.
- Submission week 7.

Decisions Made - agenda item 5

- Timeline agreed upon.
-

Agenda item 6: Backtesting Strategy

Key Discussions - agenda item 6

- Use cutoff dates (e.g., 6/18 months ago) for performance evaluation.
- Minimum of three backtests.

Decisions Made - agenda item 6

- Implement multiple cutoff date strategy.

Unresolved Issues - agenda item 6

- Visualization of classification backtest results.
-

Agenda item 7: EDA Review (Hristiyan's work) & Data Validation

Key Discussions - agenda item 7

- Engineered features presented.
- 2% events vs. 98% normal.
- Apple stock data has a large price jump.

Decisions Made - agenda item 7

- Hristiyan's EDA to be validated by another team member.
- Validation to be documented in Trello.

Unresolved Issues - agenda item 7

- Cause of Apple stock anomaly.
-

Agenda item 8: Meeting Minutes Template

Key Discussions - agenda item 8

- Word template formatting and time consumption.
- Markdown template proposed.

Decisions Made - agenda item 8

- Markdown template approved.
 - Submit as PDF, one file per meeting.
-

Agenda item 9: Task Distribution & Trello Documentation

Key Discussions - agenda item 9

- Szymon shared task plan:
 - Hristiyan: Data Engineering
 - Szymon: Team Lead, Modeling
 - Mohammadali: Supporting Models, XAI
 - Aristotelis: Dashboard/UI

Decisions Made - agenda item 9

- Task split agreed.
- Trello documentation with clear deliverables planned.

Unresolved Issues - agenda item 9

- Detailed task breakdown and progress tracking in Trello.
-

☐ Action Items

Task	Responsible	Deadline
Complete EDA	Hristiyan	End of this week
Prepare final data table	Hristiyan	Next Tuesday
Validate Hristiyan's EDA and data	Szymon (offered)	Next Tuesday
Document EDA validation in Trello	Validator	Next Tuesday
Investigate Apple stock data discrepancy	Hristiyan	Next Tuesday

Task	Responsible	Deadline
Document task distribution and plan in Trello	Szymon & team	ASAP
Figure out evidencing for missed retrospectives	Team	ASAP
Start model building	Szymon, Mohammadali, Hristiyan	After EDA/Table
Adopt Markdown template for future meeting minutes	All (Szymon to possibly lead)	Next meeting

□ Next Meeting

Date: 28-05-2025

Time: 17:00 **Location:** Teams Call

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621)

Agenda and Minutes 2-06-2025

□ Meeting Agenda – *Team 13* - 2-06-2025

Group Number: 13 **Date:** 2-06-2025 **Time:** 13:00 - 13:40 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
 - Hristiyan Georgiev (241226)
 - Mohammadali Jaber (244437)
 - Aristotelis Protopapas (234062)
-

□ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Aristotelis' Task Presentation	Aristotelis	10 min	UI/UX proposal and research
2	Hristiyan Task Progress & Presentation	Hristiyan	10 min	Feature engineering overview
3	Mohammadali's & Szymon's Tasks Presentation	Mohammadali & Szymon	10 min	Modeling results and EDA
4	Data Coverage Issue & Available Options Discussion	Szymon, All	5 min	Alternative datasets and handling missing values

#	Topic	Presenter	Duration	Notes
5	Tasks division for Week 5	All	5 min	Finalizing responsibilities

☐ Meeting Minutes – *Team 13 - 2-06-2025*

Group Number: 13 **Date:** 2-06-2025 **Time:** 13:00 - 13:40 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
 - Hristiyan Georgiev (241226) – Present
 - Mohammadali Jaber (244437) – Present
 - Aristotelis Protopapas (234062) – Present
-

☐ Discussion Summary

Agenda item 1: Aristotelis' Task Presentation

Key Discussions

- Aristotelis was asked to present the progress on his UI/UX task. He shared a short paragraph that appeared to be generated by ChatGPT and failed to provide any meaningful explanation or implementation of the design.
- The team expressed frustration over the continued lack of engagement and deliverables from Aristotelis. This was not the first time concerns were raised regarding his contributions.
- It became evident that Aristotelis had not taken ownership of his assigned task and had not delivered any actionable output. The rest of the team has covered multiple areas of the project while Aristotelis has consistently failed to provide progress or communication.
- Szymon emphasized that repeated reminders have been ignored and the team's ability to meet deadlines is being impacted.
- The situation was summarized and the team agreed that this behavior needed to be transparently reported to the mentor.

Decisions Made

- Aristotelis is required to prepare and submit an actual draft UI implementation immediately (no later than 03-06-2025).
- The team will formally raise the issue of non-contribution to the mentor to ensure accountability.
- If no improvement is observed in the upcoming days, the remaining team members may proceed without including Aristotelis in further collaborative responsibilities.

Unresolved Issues

- Uncertainty remains regarding Aristotelis' ability and willingness to contribute meaningfully to the project.

- The team must decide whether to reassign the UI task entirely if no progress is made.
-

Agenda item 2: Hristiyan Task Progress & Presentation

Key Discussions

- Hristiyan explained long- and short-term feature groups: moving averages, volume metrics, time patterns, and macro indicators.
- He discussed the challenge of handling missing values and the need for model-based testing to refine features.
- Features were derived from domain analysis and past market modeling practices.

Decisions Made

- Hristiyan will prepare a report summarizing key features and share the dataset.
- Team agreed on focusing initial modeling on “normal,” “crash,” and “rally” event types.

Unresolved Issues

- Need for more testing to determine most useful features.
-

Agenda item 3: Mohammadali's & Szymon's Tasks Presentation

Key Discussions

- Mohammadali shared results from XGBoost, decision trees, and random forest models.
- Accuracy was promising but affected by class imbalance.
- He will run random forest-based feature importance and refine long/short-term predictors.
- Szymon worked on task 2.2, validated Hristiyan's dataset, and performed EDA with mutual information heatmaps.
- Szymon introduced the idea of using Booking Holdings data due to fewer missing values.

Decisions Made

- Modeling will officially begin on 03-06-2025.
 - Szymon shared a new dataset for trials and committed to updating Trello tasks.
-

Agenda item 4: Data Coverage Issue & Available Options Discussion

Key Discussions

- Szymon contacted the mentor about gaps in the Apple dataset.
- The team may switch to Booking Holdings data or another series with fewer gaps.

- Interpolation results appeared promising for handling missing values.

Decisions Made

- Use Booking Holdings dataset as a fallback option.
- Monitor mentor's response for further guidance.

Agenda item 5: Tasks division for Week 5

Key Discussions

- Everyone should log progress on GitHub.
- Mohammadali will continue model refinement.
- Hristiyan will finalize the feature report and prepare additional macro event data.
- Aristotelis was reminded to contribute meaningfully and consistently.

Decisions Made

- All team members will upload their work to GitHub regularly.
- Trello will be updated with Week 5 task assignments.

☐ Action Items

Task	Responsible	Deadline
Deliver UI implementation draft	Aristotelis	03-06-2025
Finalize and share feature report	Hristiyan	03-06-2025 (morning)
Complete feature importance testing and short-term model	Mohammadali	03-06-2025
Upload EDA results	Szymon	03-06-2025
Update Trello and GitHub with current progress	All	03-06-2025

☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

Date: 3-06-2025 **Time:** 13:00 - 13:30 **Location:** On Campus **Chair:** Szymon Chirowski (242621) **Minutes**

Taker: Szymon Chirowski (242621)

Agenda and Minutes 04-06-2025

☐ Meeting Agenda – Team 13 - 04-06-2025

Group Number: 13 **Date:** 04/06/2025 **Time:** 13:30 – 13:45 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaberli (244437)
- Aristotelis Protopapas (234062)

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Review of Szymon's EDA	Szymon	5 min	Summarization & feedback needed
2	Modeling Responsibilities Clarification	All	2 min	Confirm tasks and split models
3	Dashboard Progress & Demo	Aristotelis	5 min	Live demo and design discussion
4	Resampling Strategy and Dataset Versions	Szymon	1 min	-
5	Task planning for next 24h	All	1 min	Clarifying work before next standup

☐ Meeting Minutes – Team 13 - 04/06/2025

Group Number: 13 **Date:** 04/06/2025 **Time:** 13:30 – 13:45 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
- Hristiyan Georgiev (241226) – Present

- Mohammadali Jaber (244437) – Present
 - Aristotelis Protopapas (234062) – Present
-

□ Discussion Summary

Agenda item 1: Review of Szymon's EDA

Key Discussions

- Szymon asked for feedback on the EDA shared via GitHub; Hristiyan reviewed it, but Mohammadali hadn't checked it yet.
- The EDA was confirmed to be solid, but a summarized interpretation of insights was requested.

Decisions Made

- Szymon will provide a summary of the EDA findings.
 - Team members will review EDA on GitHub and provide comments.
-

Agenda item 2: Modeling Responsibilities Clarification

Key Discussions

- Szymon will work on LSTM, SVM, and TFT models, as these are complex and training-intensive.
- Mohammadali confirmed he's currently working on XGBoost, Random Forest, and Decision Tree models.
- Aristotelis was assigned CNN model for time series.
- Hristiyan will work on the LSTM model as it is known to be the most time and resource-consuming, so he will assist Szymon with it.
- Each team member must document their assigned models on Trello.

Decisions Made

- Modeling assignments were confirmed:
 - Szymon: LSTM, SVM, TFT
 - Mohammadali: XGBoost, RF, DT
 - Aristotelis: CNN
 - Hristiyan: LSTM (assisting Szymon)
 - All members must log model tasks on Trello and keep it updated.
-

Agenda item 3: Dashboard Progress & Demo

Key Discussions

- Aristotelis showed a working dashboard with three pages: overview, ticker details, and prediction logs.
- Team suggested simplifying focus to one main page: ticker details + alert logs.
- Team emphasized that heatmaps and top movers are irrelevant to current scope.
- Everyone agreed the details page with prediction gauge and logs is the strongest part.
- A color palette suggestion site was shared to improve design coherence.

Decisions Made

- Aristotelis will refine dashboard based on feedback and focus on relevant event prediction data only.
- Alert log section will be improved with textual summaries following visuals.
- Deliverables must be pushed to GitHub and linked in Trello with PR links.

Unresolved Issues

- Overview page's usefulness remains questionable due to data limitations.
-

Agenda item 4: Resampling Strategy and Dataset Versions

Key Discussions

- Szymon discussed resampling of time-series data and shared graphs showing yearly closing price.
- Team discussed lack of experience with resampling but confirmed visual results looked valid.
- Mohammadali confirmed he was still working with Apple stock but would now try BKNG data.

Decisions Made

- Use resampled data where applicable and compare results to original.
 - Szymon to upload resampling notebook to GitHub for team use.
-

Agenda item 5: Task planning for next 24h

Key Discussions

- Mohammadali plans to finish three models by the end of the day.
- Discussion on what to work on next included model comparison preparation and result ranking.
- Mohammadali will create a results table showing accuracy and performance of each model.

Decisions Made

- Model ranking table will be used to decide final models for dashboard integration.

- Next standup will include model selection based on performance summaries.

□ Action Items

Task	Responsible	Deadline
Finalize EDA summary and push to GitHub	Szymon	06-06-2025
Finish XGB, RF, DT models + performance table	Mohammadali	05-06-2025
Implement CNN-based time series classification model	Aristotelis	06-06-2025
Refactor dashboard based on team feedback	Aristotelis	05-06-2025
Push resampling notebook to GitHub	Szymon	04-06-2025
Log all model/dash tasks and progress updates on Trello	All	Ongoing

□ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

□ Next Meeting

Date: 05/06/2025 **Time:** 13:00 – 13:30 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621)

Agenda and Minutes Week X

□ **Meeting Agenda – Project Update - 05/06/2025** Group Number: 13 Date: 05/06/2025 Time: 11:02 AM – 11:09 AM (approx) Location: Online Chair: Georgiev, Hristiyan (241236) Minutes Taker: Georgiev, Hristiyan (241236) (started transcription) Attendees:

- Georgiev, Hristiyan (241236)
- Chirowski, Szymon (242621)
- Jaber, Mohammadali (244437)
- Protopapas, Aristotelis (234062)

□ Agenda Items

1. **Topic:** Task Updates
 - **Presenter:** All
 - **Duration (Approx):** 7 mins
 - **Notes:** Each member to provide updates on tasks
2. **Topic:** Next Steps
 - **Presenter:** All
 - **Duration (Approx):** Ongoing
 - **Notes:** Discuss plan for the day
3. **Topic:** Other discussions

- **Presenter:** All
- **Duration (Approx):** As needed
- **Notes:** Train strike, Retrospective attendance

□ **Meeting Minutes – Project Update - 05/06/2025** Group Number: 13 Date: 05/06/2025 Time: 11:02 AM – 11:09 AM

Location: Online Chair: Georgiev, Hristiyan (241236) Minutes Taker: Georgiev, Hristiyan (241236) Attendees:

- Georgiev, Hristiyan (241236) – on time
- Chirowski, Szymon (242621) – on time
- Jaber, Mohammadali (244437) – on time
- Protopapas, Aristotelis (234062) – on time

□ **Discussion Summary**

Agenda Item	Topic	Key Discussions	Decisions Made	Unresolved Issues
1	Task Updates: Mohammadali Jaber	- Ran model distributors on data using two techniques: Smalls and different weights based on value counts in the target. - Achieved 30% accuracy with the weighting technique.	- Continue working with the weighting technique. - Change some parameters to get better accuracy.	- How to improve accuracy further (currently 30%). - Clarification on target accuracy (25% vs 51%).
1	Task Updates: Aristotelis Protopapas	- Has not started building a CNN model yet, plans to start today. - Model will likely be a classification model.	- Will ask for help if needed. - Will work with Ali as they are working on similar models.	- Needs to start and complete the CNN model.
1	Task Updates: Szymon Chirowski	- Worked on fixing environment. - Calculated baselines: ZeroR (51-52%), Weighted Guess, Persistent model (6% accuracy). - Current validation accuracy is 8%.	- Passed the baseline block with 8% accuracy.	- Environment issues.
1	Task Updates: Hristiyan Georgiev	- Decided to combine two models: CNN + GRU (Gated Recurrent Unit, clarified from "GP"). - Fixed model architecture. - Testing was terrible and took 90 minutes.	- Needs to fix environment for faster runs. - Needs to adjust the model.	- Environment issues impacting speed. - Model performance needs significant improvement.

Agenda Item	Topic	Key Discussions	Decisions Made	Unresolved Issues
2	Next Steps	- All to continue with their current tasks. - Aristotelis to work with Ali on his model and ask questions in the group chat if needed. - Mohammadali might try a Time Series Transformer (TST) model.	- Proceed with individual tasks.	
3	Other Discussions: Train Strike	- Aristotelis mentioned a train strike for the next day. - Teacher addressed the issue: option to join the retrospective online if unable to be in school.	- Aristotelis will check transport options and inform the team. - Members can attend the retrospective online if affected by the strike.	- Certainty of Aristotelis's attendance in person for the retrospective.

□ **Action Items**

Task	Responsible	Deadline	Status
Continue improving accuracy of the weighting model	Jaberi, Mohammadali (244437)	Ongoing	In Progress
Start and build CNN classification model	Protopapas, Aristotelis (234062)	Today	To Start
Collaborate with Ali on CNN model development	Protopapas, Aristotelis (234062)	Ongoing	To Start
Fix environment and adjust combined CNN+GRU model	Georgiev, Hristiyan (241236)	Ongoing	In Progress
Continue model development and environment fixes	Chirowski, Szymon (242621)	Ongoing	In Progress
(Optional) Explore Time Series Transformer (TST) model	Jaberi, Mohammadali (244437)	Ongoing	To Evaluate
Check transport for retrospective & inform team	Protopapas, Aristotelis (234062)	By EOD	To Do

□ Make sure all tasks and deadlines are also logged in Trello or your task manager.

□ **Next Meeting** Date: [Not explicitly stated, assumed next regular meeting] Time: [Not explicitly stated] Location: [Not explicitly stated, likely Online] Chair: [Not explicitly stated, likely rotating] Minutes Taker: [Not explicitly stated, likely rotating]

Agenda and Minutes

06/06/2025

☐ Meeting Agenda – Team 13 - 06/06/2025

Group Number: 13 **Date:** 06/06/2025 **Time:** 10:00 – 10:30 **Location:** On Campus and Teams Call **Chair:**
Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaber (244437)
- Aristotelis Protopapas (234062) - Online (train strike)
- Elavendan Rajendran

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Progress on modeling tasks and hybrid model	Szymon, Mohammadali, Hristiyan	10 min	Updates on LSTM, hybrid CNN models, and RF/DT
2	CNN model progress & UI status update	Aristotelis	10 min	Clarification of CNN progress and dashboard alignment
3	Clarifications on SQL tasks and template questions	Szymon, Elavendan	10 min	Addressing doubts about SQL security and final notebook

☐ Meeting Minutes – Team 13 - 06/06/2025

Group Number: 13 **Date:** 06/06/2025 **Time:** 10:00 – 10:30 **Location:** On Campus and Teams Call **Chair:**
Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) - Present
- Hristiyan Georgiev (241226) - Present
- Mohammadali Jaber (244437) - Present

- Aristotelis Protopapas (234062) - Present
 - Elavendan Rajendran - Present
-

□ Discussion Summary

Agenda item 1: Progress on modeling tasks and hybrid model

Key Discussions - agenda item 1

- Hristiyan shared progress on the hybrid CNN-LSTM model.
- Szymon showed his progress with LSTM model, initial runs were time-consuming due to environment issues, but promising (7% validation accuracy vs 0% baseline).
- Mohammadali reported success with RF and DT models using class weighting; best F1 score ~30%.
- Discussion on preprocessing: Hristiyan clarified that main preprocessing (target shift, scaling) was already completed and shared in the repo.

Decisions Made - agenda item 1

- Szymon to continue hybrid model optimization and address environment issues.
 - Mohammadali to finalize performance comparisons for all models.
 - Team agreed to prioritize model performance results before week 7.
-

Agenda item 2: CNN model progress & UI status update

Key Discussions - agenda item 2

- Aristotelis reported limited progress: only basic preprocessing done despite team preprocessing already shared.
- Dashboard demo showed 3 pages with dummy data; concerns raised about relevance of the charts and unclear linkage to project goals.
- Team and mentor emphasized urgent need for Aristotelis to align with project objectives, integrate team models into dashboard, and stop pursuing unrelated ideas (live API).
- Mentor issued a second formal warning regarding contribution quality and deadline risk (possible group contribution = 0 if no meaningful work by next Tuesday).

Decisions Made - agenda item 2

- Aristotelis must:
 - Complete CNN model and show results.
 - Refactor dashboard to match project scope and integrate proper predictions.
 - Follow Git workflow (branch + PR) and contribute properly to the repo.
- Other members may take over UI if no progress is made by next week.

Unresolved Issues

- Dashboard currently not usable as-is; urgent rework needed.
- CNN model not yet delivered.

Agenda item 3: Clarifications on SQL tasks and template questions

Key Discussions - agenda item 3

- Szymon asked about "Create table for queries" — clarified this refers to saving query results in a new table.
- Security questions: no need to implement; just document role-based access suggestions in Markdown.
- Template questions: how to document preprocessing (OK to call existing functions), what is expected in final step (model output for UI).

Decisions Made - agenda item 3

- Szymon to proceed as discussed and update To Do list accordingly.
- No blockers identified for SQL work or template completion.

☐ Action Items

Task	Responsible	Deadline
Continue optimizing hybrid CNN-LSTM model	Szymon	Ongoing
Finalize RF/DT/XGB model performance results	Mohammadali	Ongoing
Implement CNN model and refactor dashboard per team feedback	Aristotelis	11-06-2025
Review and sign off peer-reviewed SQL and EDA	All	ASAP

☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

Date: 10/06/2025 **Time:** 10:00 - 10:30 **Location:** On Campus and Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621)

Agenda and Minutes 10/06/25

☐ Meeting Agenda – Team 13 - 10/06/25

Group Number: 13 **Date:** 10/06/25 **Time:** 10:00 - 10:30 **Location:** On Campus & Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaber (244437)
- Aristotelis Protopapas (234062) – Present (Online, train strike)
- Elavendan Rajendran

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Review of CNN model & classification	Aristotelis	10 min	Team feedback on issues with the CNN model
2	Code clarity & use of frameworks	All	10 min	Misunderstandings around PyTorch implementation
3	Final decision on UI task ownership	Mentor	10 min	Evaluation of contribution and group direction

☐ Meeting Minutes – *Team 13 - 10/06/25*

Group Number: 13 **Date:** 10/06/25 **Time:** 10:00 - 10:30 **Location:** On Campus & Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
- Hristiyan Georgiev (241226) – Present
- Mohammadali Jaber (244437) – Present
- Aristotelis Protopapas (234062) – Present (Online)
- Elavendan Rajendran – Present

☐ Discussion Summary

Agenda item 1: Review of CNN model & classification

Key Discussions

- Aristotelis shared a classification report showing 0% performance on the majority class, raising immediate concerns.
- Szymon and Hristiyan pointed out potential issues with label mismatch or incorrect preprocessing, but Aristotelis couldn't explain the architecture or results clearly.
- The team emphasized the importance of being able to defend implementation choices and provide reproducible results.

Decisions Made

- Aristotelis was asked to upload the complete and testable version of the CNN model.
 - The model was marked as unusable in its current form, pending independent review by other team members.
-

Agenda item 2: Code clarity & use of frameworks

Key Discussions

- Aristotelis was asked which framework he used; he couldn't confirm PyTorch usage despite saving a .pt file.
- The team expressed loss of trust in the validity of his code due to repeated inability to answer technical questions or explain his work.
- Mentor intervened, stating that after multiple warnings, the behavior was unacceptable.

Decisions Made

- The team will independently review the code, without relying on Aristotelis' clarification.
 - Aristotelis was formally declared "not reliable" for remaining technical contributions by the mentor.
-

Agenda item 3: Final decision on UI task ownership

Key Discussions

- Mentor reviewed Aristotelis' overall performance and lack of contribution across previous weeks.
- The team confirmed that he had not integrated team feedback or produced working components.
- Mentor explicitly stated that Aristotelis' **group contribution score was set to 0** due to repeated failure to contribute meaningfully.
- Going forward, the team is allowed—and encouraged—to continue project development without involving Aristotelis in any way.

Decisions Made

- UI task is now fully reassigned to other group members (Hristiyan).
 - Aristotelis will not be included in further planning, development, or integration of the project.
 - Team is officially released from coordination responsibilities with Aristotelis.
-

□ Action Items

Task	Responsible	Deadline
Finalize CNN model analysis & discard if needed	Szymon, Hristiyan	11-06-2025
Rebuild and polish dashboard UI	Hristiyan	12-06-2025
Update documentation to reflect team structure	All	ASAP

Task	Responsible	Deadline
Upload final model results	Mohammadali	12-06-2025

☐ Make sure all tasks and deadlines are also logged in Trello or your task manager.

☐ Next Meeting

Date: 11/06/25 **Time:** 20:00 - 20:30 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621)

Agenda and Minutes 11/06/25

☐ Meeting Agenda – Team 13 - 11/06/25

Group Number: 13 **Date:** 11/06/25 **Time:** 20:00–20:30 **Location:** Teams Call **Chair:** Szymon Chirowski (242621)
Minutes Taker: Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaber (244437)
- Aristotelis Protopapas (234062)

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	UI and deployment discussion	Hristiyan	10 min	UI stack, deployment integration with ML pipeline
2	Final ML pipeline and modularization plan	Szymon	10 min	Project structure and API interface
3	Hyperparameter tuning & code formatting	Mohammadali	10 min	Prioritizing tuning over new model, logging standards

☐ Meeting Minutes – Team 13 - 11/06/25

Group Number: 13 **Date:** 11/06/25 **Time:** 20:00–20:50 **Location:** Teams Call **Chair:** Szymon Chirowski (242621)
Minutes Taker: Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
- Hristiyan Georgiev (241226) – Present

- Mohammadali Jaberi (244437) – Present
 - Aristotelis Protopapas (234062) – Absent (No show, no prior notice)
-

□ Discussion Summary

Agenda item 1: UI and deployment discussion

Key Discussions

- Hristiyan will deliver a functional dashboard by Friday using Flask and HTML.
- UI will connect to saved models for inference through an endpoint.
- Szymon and Hristiyan discussed the integration of prediction logic through Flask routing.
- Clarified UI doesn't need to support model training, only prediction.

Decisions Made

- Hristiyan will continue with Flask due to prior experience.
 - Focus will remain on deploying working prediction flow rather than visual polish.
-

Agenda item 2: Final ML pipeline and modularization plan

Key Discussions

- Szymon proposed organizing the repo into 3-4 main modules: preprocessing, modeling, dashboard, and optionally training.
- Manual prediction scripts will be integrated through endpoint calls.
- Confirmed only one final model needs to be included for submission.

Decisions Made

- Repo structure will include clearly separated modules for preprocessing, training, and prediction.
 - Final model and artifacts will be included as static files.
 - Model training logic will not be exposed in the dashboard interface.
-

Agenda item 3: Hyperparameter tuning & code formatting

Key Discussions

- Team decided to focus on hyperparameter tuning rather than adding a new model (TFT).
- Mohammadali will use manual search or cross-validation for tuning.
- Logging will replace print statements for MLOps formatting compliance.

Decisions Made

- Mohammadali to perform hyperparameter tuning and document progress in Trello.
- Logging setup will follow template configuration provided by Szymon.
- Codebase will be formatted with Black; front-end is excluded from formatting standards.

□ Action Items

Task	Responsible	Deadline
Finalize UI and prediction endpoint with Flask	Hristiyan	14/06/25
Hyperparameter tuning using manual or CV methods	Mohammadali	14/06/25
Repo modularization (training, prediction)	Szymon	14/06/25
Apply Black formatting and logging configuration	All	14/06/25
Add poetry environment setup and screenshots	All	14/06/25
Start README draft (structure, decisions, results)	All	17/06/25

□ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

□ Next Meeting

Date: 12/06/25 **Time:** 13:00–13:15 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621)

Agenda and Minutes 12/06/25

□ Meeting Agenda – Team 13 - 12/06/25

Group Number: 13 **Date:** 12/06/25 **Time:** 13:00–13:30 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaber (244437)
- Aristotelis Protopapas (234062)

□ Agenda Items

#	Topic	Presenter	Duration	Notes
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#	Topic	Presenter	Duration	Notes
1	Work progress review (past 12 hours)	Team	10 min	Tasks completed since previous meeting
2	Next 24h task planning	Team	10 min	Trello alignment and planning
3	Roadblocks	Team	10 min	Addressing blockers for each team member

□ Meeting Minutes – *Team 13 - 12/06/25*

Group Number: 13 **Date:** 12/06/25 **Time:** 13:00–13:10 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621) **Attendees:**

- Szymon Chirowski (242621) – Present
 - Hristiyan Georgiev (241226) – Present
 - Mohammadali Jaberli (244437) – Present
 - Aristotelis Protopapas (234062) – Absent (No show, no prior notice)
-

□ Discussion Summary

Agenda item 1: Work progress review (past 12 hours)

Key Discussions

- Szymon added docstrings and made minor updates to his code.
- Hristiyan built the UI, which is currently functional and ready for presentation; further polishing will be done next week.
- Mohammadali started hyperparameter tuning on the LSTM and finished setting up the poetry environment.

Decisions Made

- Work on the UI, tuning, and final engineering code should proceed without delays.
 - All code must be pushed to GitHub and branches synchronized.
-

Agenda item 2: Next 24h task planning

Key Discussions

- Szymon will handle backend integration and finalize the poetry setup so others can easily reproduce the environment.
- Hristiyan committed to working through his Trello tasks including refactoring the engineering script and uploading the UI.

- Mohammadali will continue hyperparameter tuning and work on his learning log.

Decisions Made

- All members must update Trello progress today.
- UI must be merged to `dev` branch ASAP to unblock backend work.

Agenda item 3: Roadblocks

Key Discussions

- Szymon's only blocker was waiting for the UI branch to be pushed to `dev`.
- Hristiyan was already working on resolving that blocker by creating a merge-ready branch.
- Mohammadali confirmed no current blockers.

Decisions Made

- Merge UI immediately to remove bottlenecks.
- If no major errors are encountered, next steps can proceed as planned.

☐ Action Items

Task	Responsible	Deadline
Merge UI branch into <code>dev</code>	Hristiyan	ASAP
Hyperparameter tuning	Mohammadali	13/06/25
Refactor and upload engineering script	Hristiyan	13/06/25
Update Trello and GitHub repositories	All	13/06/25

- ☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

Date: 13/06/25 **Time:** 10:00–10:30 **Location:** On Campus **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Szymon Chirowski (242621)

Agenda and Minutes 16/06/25

☐ Meeting Agenda – Team 13 - 16/06/25

Group Number: 13 **Date:** 16/06/25 **Time:** 10:59 – 11:17 **Location:** Teams Call

Chair: Szymon Chirowski (242621)

Minutes Taker: Hristiyan Georgiev (241236) **Attendees:**

- Szymon Chirowski (242621)
 - Hristiyan Georgiev (241236)
 - Mohammadali Jaberli (244437)
 - Aristotelis Protopapas (234062)
-

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Task assignment for the final sprint	Szymon Chirowski	17 mins	Assigning all remaining tasks with a deadline of Thursday midday.
2	README file creation and content	Szymon Chirowski		Discussion on the scope and content of the project's README file.
3	MLOps tasks and Final Presentation	Szymon Chirowski		Covering unit tests [cite: 49], application deployment [cite: 51], and the final presentation slides.
4	Final Project Template and blocker discussion	Szymon Chirowski		Discussion on completing the project template and issues with hyperparameter tuning.

☐ Meeting Minutes – *Team 13 - 16/06/25*

Group Number: 13 **Date:** 16/06/25 **Time:** 10:59 – 11:17 **Location:** Teams Call

Chair: Szymon Chirowski (242621)

Minutes Taker: Hristiyan Georgiev (241236) **Attendees:**

- Szymon Chirowski (242621) - Present
 - Hristiyan Georgiev (241236) - Present
 - Mohammadali Jaberli (244437) - Present
 - Aristotelis Protopapas (234062) - Absent (No Show)
-

☐ Discussion Summary

Agenda item 1: Task assignment for the final sprint

Key Discussions - agenda item 1

- Szymon introduced the plan to assign all remaining tasks for the next three days.
- The goal is to have everything completed by midday on Thursday, leaving the rest of Thursday and Friday free.
- Szymon had already created cards for the tasks in Trello to facilitate the discussion.

Agenda item 2: README file creation

Key Discussions - agenda item 2

- The team discussed the creation of a single, comprehensive README file for the project.
- Hristiyan initially questioned the content for the README, suggesting it usually contains setup instructions.
- Szymon clarified that the README should include details about data modeling decisions, key transformations, and database creation, as detailed in the Trello card.
- Hristiyan agreed and suggested he could use the README to go into more detail about why certain data modeling choices were made, which wasn't covered in his paper.

Decisions Made - agenda item 2

- Hristiyan Georgiev will be responsible for writing the project README file.
- The README will include justifications for data modeling choices.

Agenda item 3: MLOps tasks and Final Presentation

Key Discussions - agenda item 3

- Szymon requested unit tests for the data preprocessing from Hristiyan.
- Hristiyan is to either deploy the demo application or provide screenshots of it running locally.
- Szymon pointed out that the application needs a banner to indicate that predictions are AI-generated.
- The team discussed creating unit tests for the Flask back-end, but Hristiyan noted he might need to talk with a teacher about the necessity of this task due to time constraints.
- Mohammadali agreed to create the final presentation and will share it with the team for feedback.

Decisions Made - agenda item 3

- Hristiyan will provide the unit tests for data preprocessing.
- Hristiyan will deploy the demo or provide screenshots.
- Hristiyan will add an "AI only" banner to the application's UI.
- Mohammadali will prepare the final presentation and circulate it for feedback.
- A decision on who will write the back-end unit tests (Szymon or Mohammadali) will be made the next day.

Agenda item 4: Final Project Template and blocker discussion

Key Discussions - agenda item 4

- Szymon will take responsibility for combining all the project files into a single directory and generating the documentation.

- A significant issue was raised regarding the hyperparameter tuning task for the final project template.
- Szymon expressed concern that this task requires training a new model, which is time-consuming (20 minutes per training) and the current model's predictions are "questionable".
- The team agreed that they need guidance on how to proceed with this task.

Decisions Made - agenda item 4

- Szymon will combine all project components and handle documentation.
- The team will discuss the hyperparameter tuning issue with their teacher during the next in-person meeting.

☐ Action Items

Task	Responsible	Deadline
Write the project README file	Hristiyan Georgiev	Thursday (19/06/25) Midday
Prepare the final presentation	Mohammadali Jaber	Thursday (19/06/25) Midday
Provide unit tests for data preprocessing	Hristiyan Georgiev	Thursday (19/06/25) Midday
Deploy demo application or provide screenshots	Hristiyan Georgiev	Thursday (19/06/25) Midday
Add AI disclosure banner to the app	Hristiyan Georgiev	Thursday (19/06/25) Midday
Combine all project files into one directory	Szymon Chirowski	17/06/25
Generate project documentation	Szymon Chirowski	Thursday (19/06/25) Midday
Fill in final project template	Szymon Chirowski	Thursday (19/06/25) Midday
Discuss hyperparameter tuning with the teacher	All members	17/06/25

☐ Make sure all tasks and deadlines are also logged in Trello or your task manager.

☐ Next Meeting

Date: 17/06/25 **Time:** [10:00 - 10:30]

Location: On Campus **Chair:** Szymon Chirowski (242621)

Minutes Taker: Hristiyan Georgiev (241236)

Agenda and Minutes 17/06/25

☐ Meeting Agenda – *Team 13 - 17/06/25*

Group Number: 13 **Date:** 17/06/25

Time: 13:18 – 13:52

Location: On Campus

Chair: Szymon Chirowski (242621)

Minutes Taker: Hristiyan Georgiev (241226) **Attendees:**

- Szymon Chirowski (242621)
 - Hristiyan Georgiev (241226)
 - Mohammadali Jaberli (244437)
 - Aristotelis Protopapas (234062)
-

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Icebreaker: The Project as a Movie	Team	5 mins	Each team member described the project as if it were a movie, including its genre and title.
2	Project Retrospective	Team	20 mins	The team used sticky notes to document what went well, what didn't go well, and potential improvements for the project and the block.
3	Kudos and Compliments	Team	5 mins	Team members wrote and shared compliments for each other to end the project on a positive note.

☐ Meeting Minutes – *Team 13 - 17/06/25*

Group Number: 13 **Date:** 17/06/25

Time: 13:18 – 13:52

Location: On Campus

Chair: Szymon Chirowski (242621)

Minutes Taker: Hristiyan Georgiev (241226) **Attendees:**

- Szymon Chirowski (242621) - Present
 - Hristiyan Georgiev (241226) - Present
 - Mohammadali Jaberli (244437) - Present
 - Aristotelis Protopapas (234062) - Absent
-

☐ Discussion Summary

Agenda item 1: The Project as a Movie (Icebreaker)

Key Discussions - agenda item 1

- **Hris:** Described the project as a comedy or a "reality show for data scientists". This was because they were trying to simulate a real-world environment but, as a school project, it wasn't fully functional. Errors were not taken too seriously, and some issues that would be critical in a real job were acceptable in an academic setting.
- **Szymon:** Labeled the project as an "action/soap opera". "Action" because something was always happening, and "soap opera" due to the block feeling disorganized. A major issue was an uncooperative teammate, which required the rest of the team to cover his tasks.
- **Ali:** Titled the movie "Living in the next 30 minutes" with a genre of science fiction. This was because while the project's predictive goals weren't currently possible, they might be in the future with the right data, like news feeds.

Decisions Made - agenda item 1

- This was an icebreaker exercise; no formal decisions were made.
-

Agenda item 2: Project Retrospective

Key Discussions - agenda item 2

What Went Well:

- **Teamwork:** The team felt they integrated well and shared a similar vision. Dealing with a non-contributing member was seen as a realistic experience that brought the rest of the team closer.
- **Idea Generation:** The team's ideas were very similar from the start, allowing them to finalize their project concept in a single day.
- **Feature Engineering:** Hris was praised for quickly adapting the feature engineering when the team had to switch from Apple to P&G stock due to missing data. The features were considered good and helpful for the model.
- **Model Implementation:** The team successfully implemented different models, including an LSTM model, and achieved a "usable" result after hyperparameter tuning.

What Didn't Go Well:

- **Block Structure:** The project started too late (week 3 or 4), compressing a significant amount of work into the final week. It was noted the block was shortened from a planned 8 weeks to 7.
- **Initial Data Quality:** The first dataset had significant data gaps, an issue that was discovered late in week 5 because the person responsible had not communicated the problem.
- **UI Development:** Too much time was spent on UI tasks (HTML) instead of core data science work. This was exacerbated by trusting the task to a teammate who did not complete it, forcing a last-minute effort from others.
- **Model Performance:** Szymon expressed dissatisfaction with the final model, stating it was suitable for academia but not for a real-world environment. This was largely because it lacked external data like news, which is crucial for stock market prediction.
- **Hyperparameter Tuning Task:** The team was confused about how to approach the hyperparameter tuning task, as they felt they had already performed extensive tuning to arrive at their current model and creating another

model didn't make sense.

Improvements:

- **Earlier Technical Training:** The team suggested that training on essential tools like Git, GitHub, and concepts like Poetry testing should be introduced in Week 1, not Week 6.
- **Better Teamwork Workshops:** The mandatory Lego teamwork workshop was seen as ineffective and restrictive. It was suggested that workshops should focus on practical teamwork skills like task division.

Decisions Made - agenda item 2

- The team agreed that the feedback on the block's structure and content should be formally submitted through feedback forms.
- The team decided to ask a teacher for clarification on the hyperparameter tuning task.

Agenda item 3: Kudos and Compliments

Key Discussions - agenda item 3

- The team concluded the meeting with a positive exercise, writing and sharing compliments on sticky notes.
- Hristiyan was complimented for his positive attitude and for working hard to complete the engineering and UI parts, even if it meant losing sleep.
- Szymon was praised for his excellent project management and leadership skills, ensuring the team stayed on track with all tasks.
- The overall sentiment was positive, with team members expressing happiness with the group and their ability to work together.

Decisions Made - agenda item 3

- This was a team-building exercise; no formal decisions were made.

☐ Action Items

Task	Responsible	Deadline
Submit official feedback regarding the block's structure and content	All members	End of Block
Seek clarification from a teacher about the hyperparameter tuning task	Team	ASAP

☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

Date: [18/06/25]

Time: [10:00 - 10:30]

Location: Teams Call

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621)

Agenda and Minutes 18/06/25

☐ Meeting Agenda – *Team 13 - 18/06/25*

Group Number: 13 **Date:** 18/06/25 **Time:** 11:00 - 11:23 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Hristiyan Georgiev (241226) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaberli (244437)
- Aristotelis Protopapas (234062)

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Final Presentation Review	Mohammadali Jaberli	15 mins	Reviewing the draft of the final presentation and providing collective feedback.
2	Individual & Group Task Status Update	All	5 mins	Each member reports on their completed tasks and remaining work.
3	Plan for Absent Teammate	All	3 mins	Discussing how to handle the presentation part and tasks of the unresponsive team member.

☐ Meeting Minutes – *Team 13 - 18/06/25*

Group Number: 13 **Date:** 18/06/25 **Time:** 11:00 - 11:23 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Hristiyan Georgiev (241226) **Attendees:**

- Szymon Chirowski (242621) - Present
- Hristiyan Georgiev (241226) - Present
- Mohammadali Jaberli (244437) - Present
- Aristotelis Protopapas (234062) - Absent

□ Discussion Summary

Agenda item 1: Final Presentation Review

Key Discussions - agenda item 1

- Mohammadali shared the initial draft of the final presentation. The structure included an introduction, company background, data sources, engineering, EDA, UI/UX, and model results.
- The team provided feedback on the slides:
 - **Title Slide:** It was suggested to add the names and student numbers of all team members for a professional look. The subtitle should be changed from "data price" to "stock data" for clarity.
 - **Text-Heavy Slides:** The team agreed that slides with large blocks of text should be summarized into bullet points. The presenter can elaborate on the points verbally.
 - **Content Slides:** The slide showing raw code should be removed, as it's not practical for a presentation. The heatmap on the EDA slide was deemed unreadable and should be replaced with a few bullet points highlighting the key findings.
 - **Concluding Slide:** The presentation should end with a "Questions?" slide rather than a "Thank you" slide.
- For the EU AI Act compliance slide, Szymon suggested summarizing the key points into three or four bullets: the project's risk level, the specific articles to comply with (Article 50 and 4), and how compliance was achieved.

Decisions Made - agenda item 1

- Mohammadali will update the presentation slides based on the team's feedback.
- Hristiyan will edit the slides related to his work (Data Engineering and UI) directly once the presentation is shared via Google Docs.
- Mohammadali will finish the draft within a few hours and share it with the team for final review.

Agenda item 2: Individual & Group Task Status Update

Key Discussions - agenda item 2

- **Szymon:** Reported that he had completed all testing for the project, achieving 92% code coverage. He noted using AI assistance for writing some challenging tests, especially for Flask, which was previously untaught. His remaining work includes documenting the last two meetings, writing the project README, and completing the final project template, all due by tonight or tomorrow morning.
- **Hristiyan:** Stated his only remaining group task is to add back-testing to the project template. He planned to complete his individual work logs by the end of the day.
- **Mohammadali:** His main outstanding individual task is to update his learning and work logs from week 3 onwards. He intended to finish the final presentation within two to three hours.

Decisions Made - agenda item 2

- The team confirmed their individual timelines and agreed to complete all remaining tasks by the following day at the latest.

Agenda item 3: Plan for Absent Teammate

Key Discussions - agenda item 3

- The team discussed the issue of their fourth member, Aristotelis, being unresponsive and how to manage his assigned presentation part (UI/UX).
- During the meeting, Szymon attempted to call Aristotelis, but the call was not answered.
- The team debated how to divide the presentation. Szymon suggested a dynamic, interwoven presentation style like a guest lecture they had seen, but Hristiyan felt the team lacked the practice for such a format and recommended sticking to divided sections.

Decisions Made - agenda item 3

- Szymon will contact their mentor, Margot, to inform her about the unresponsive team member.
- Hristiyan will prepare the UI/UX part of the presentation and edit the corresponding slides himself, so the team is ready to present without Aristotelis if necessary.
- The presentation will be divided by section, with each member presenting their respective parts.

☐ Action Items

Task	Responsible	Deadline
Revise and share the final presentation draft	Mohammadali Jaber	18/06/25
Edit Data Engineering and UI/UX slides	Hristiyan Georgiev	19/06/25
Add back-testing to the final project template	Hristiyan Georgiev	19/06/25
Complete README and final meeting documentation	Szymon Chirowski	19/06/25
Inform mentor about the unresponsive team member	Szymon Chirowski	18/06/25
Complete all individual learning and work logs	All	19/06/25
Fill out the peer feedback form	All	18/06/25, 17:00

☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

Date: [19/06/25]

Time: [13:00 - 13:30]

Location: Teams Call

Chair: Szymon Chirowski (242621)

Minutes Taker: Szymon Chirowski (242621)

Agenda and Minutes 19/06/25

☐ Meeting Agenda – Team 13 - 19/06/25

Group Number: 13 **Date:** 19/06/25 **Time:** 11:03 - 11:18 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Hristiyan Georgiev (241226) **Attendees:**

- Szymon Chirowski (242621)
- Hristiyan Georgiev (241226)
- Mohammadali Jaberli (244437)
- Aristotelis Protopapas (234062)

☐ Agenda Items

#	Topic	Presenter	Duration	Notes
1	Contribution Table Review	Szymon Chirowski	8 mins	Discussing and finalizing the point distribution for each team member across all tasks.
2	Final Task Updates	All	3 mins	Each member provides a final status update on their remaining individual and group tasks.
3	Presentation Speaker Roles	All	2 mins	Assigning specific slides to each team member for the final presentation.
4	Plan for Absent Teammate	All	2 mins	Deciding on the course of action regarding the unresponsive team member for the final presentation.

☐ Meeting Minutes – Team 13 - 19/06/25

Group Number: 13 **Date:** 19/06/25 **Time:** 11:03 - 11:18 **Location:** Teams Call **Chair:** Szymon Chirowski (242621) **Minutes Taker:** Hristiyan Georgiev (241226) **Attendees:**

- Szymon Chirowski (242621) - Present
 - Hristiyan Georgiev (241226) - Present
 - Mohammadali Jaberli (244437) - Present
 - Aristotelis Protopapas (234062) - Absent
-

□ Discussion Summary

Agenda item 1: Contribution Table Review

Key Discussions - agenda item 1

- Szymon presented a draft of the contribution distribution table for the team to review and finalize.
- **Legal Risk Assessment (Task 2.2):** Hristiyan requested and was given 10 points for his work on the UI to comply with Article 50.
- **Trello Board Management:** Szymon assigned himself 50% of the points for setting up and managing the Trello board, with the other two members receiving 25% each, which the team agreed was fair.
- **Unit Tests & Documentation:** Points were split, with Szymon and Hristiyan receiving a larger share as Ali had not contributed docstrings to his code.
- **User-Friendly Application:** The points were distributed 85% to Hristiyan for development, and 10% and 5% to the others for review.
- **Brainstorming & Innovation:** Szymon added this category after consulting a teacher, acknowledging it was a major part of the project. Points were distributed to credit Hristiyan for the thinking behind the data preprocessing and to Ali for his work on hyperparameter tuning and exploring different models.

Decisions Made - agenda item 1

- The team collectively reviewed and agreed upon the final distribution of points in the contribution table.

Agenda item 2: Final Task Updates

Key Discussions - agenda item 2

- **Szymon:** Is currently working on the project README and the final project template. He also needs to complete the documentation for the last two meetings. He expects to have all his tasks finished by the end of the day or the next morning at the latest.
- **Hristiyan:** His remaining task is to add bug testing to the final template. He located the necessary files in the GitHub repository after a brief search.
- **Mohammadali:** Confirmed he had prepared a personal script for his presentation parts.

Decisions Made - agenda item 2

- All members confirmed they are on track to complete their remaining tasks before the final presentation.

Agenda item 3: Finalizing Presentation Speaker Roles

Key Discussions - agenda item 3

- The team discussed the division of slides for the final presentation to ensure a smooth delivery.

- The agreed-upon order of topics (Engineering -> EDA -> ML Model -> UI) was considered logical as it followed the project's workflow.

Decisions Made - agenda item 3

- **Mohammadali:** Will present slides 1-6 (Introduction, project plan) and slide 8 (EDA).
- **Hristiyan:** Will present slide 7 (Data Engineering) and slide 11 (UI/UX Demo).
- **Szymon:** Will present slides 9 and 10 (Machine Learning Model and Results).

Agenda item 4: Plan for Absent Teammate

Key Discussions - agenda item 4

- The team discussed how to proceed with the final presentation given that their fourth member, Aristotelis, had been unresponsive to texts and did not show up for the meeting.

Decisions Made - agenda item 4

- The team will proceed with the presentation as a group of three.
- They decided that if the absent member appears at the final presentation without preparation, they will inform the teachers of the situation, as it will be evident he has not participated in the final stages of the project.

☐ Action Items

Task	Responsible	Deadline
Complete project README and final project template	Szymon Chirowski	19/06/25
Add bug testing to the final template	Hristiyan Georgiev	19/06/25
Prepare individual presentation parts and scripts	All	20/06/25
Finalize and submit the contribution table	Szymon Chirowski	19/06/25
Complete missing meeting documentation	Szymon Chirowski	19/06/25

- ☐ *Make sure all tasks and deadlines are also logged in Trello or your task manager.*

☐ Next Meeting

N/A - End of the block
